

# ChristosHadjinikolis

ML Tech Lead (Staff-Level) @ Vortexa Ltd | Ph.D.

Committee Expert Member @ JTC'21

✉ christos.hadjinikolis@gmail.com

## mediums

🌐 personal website

 LinkedIn

 github

 Google Scholar

 YouTube

## languages

bilingual

Greek/English

## programming

Java/Kotlin, Python,

SQL, Cypher

## author

pypi:dynamicio

## developer skills

distributed ML systems,  
CI/CD, IaC,  
observability,  
model/data versioning,  
production monitoring

## cloud-platforms

AWS, GCP

## technologies & frameworks

py/Spark, py/Flink,  
Kafka, FastApi, Docker,  
MLflow, Airflow,  
Elasticsearch

## python libs

pandas, numpy, polars

## visualisations

Gephi, Seaborn,  
plotly/streamlit

## interests

**Sports:** Tennis, Jogging,

**Reading:** Comics,  
Philosophy, Politics

## about me

Staff-level ML systems engineer with 15+ years of experience designing and operating production AI systems across data platforms, backend engineering, and platform delivery. Recent work has focused on destination-prediction services, real-time inference pipelines, streaming data infrastructure, and the operational discipline required to move models from experimentation into reliable production. Strong record of owning ambiguous, high-scale problems end-to-end, partnering closely with scientists and engineers, and building observable, maintainable systems around heterogeneous data.

## work-experience

**Vortexa Ltd, London, UK** 12/2020–present

**ML Tech Lead (Staff-Level) | Pod Lead**

Lead the design and delivery of large-scale ML systems for maritime forecasting, combining destination prediction, streaming infrastructure, backend services, and production rollout discipline. Own technical direction across cross-functional teams and drive the conversion of advanced modelling work into reliable, user-facing systems with measurable operating quality.

- **Data Platform Foundations:** Built distributed Flink and pandas workflows over geospatial, textual, and graph-structured datasets, strengthening dataflow foundations and reducing maintenance effort for downstream training and inference.
- **Production Inference Systems:** Delivered backend model-serving services with versioning, switch-over logic, auditability, and operational safeguards for continuously running prediction systems.
- **Model Lifecycle Engineering:** Built training and inference workflows with CI/CD, online evaluation, and traceable performance monitoring to support dependable deployment over time.
- **Streaming Infrastructure:** Designed Kafka-based pipelines supporting near real-time model refresh and downstream triggers across 15+ ETL components.
- **Staff-Level Ownership:** Led delivery across ML, DS, and DE functions, set engineering standards, and raised execution quality for systems moving from research-grade ideas to production behaviour.

**DataReply, London, UK** 04/2016–12/2020

**Data Scientist | ML Engineer**

Progressed from Data Scientist to ML Engineer, leading cross-functional DS/DE teams and delivering production-grade systems across UBS, Vodafone, and CNHi. Work centered on platformisation, observability, deployment workflows, and turning data-science prototypes into maintainable software systems.

- **Vodafone Group (09/2019–11/2020, London):**
  - Worked on Vodafone's internal MLOps platform (**Infinity Project**) on GCP using **Kubeflow**, supporting model versioning, deployment, telemetry, and resource-governed notebooks.
  - Implemented CI/CD pipelines, shared feature and data workflows, and monitoring for reproducible, production-ready ML systems.
- **CNHi (05/2019–09/2019, Chicago):**
  - Built **time-series forecasting** models over connected-vehicle telemetry and helped transition statistical prototypes into scalable, cloud-ready services.
- **UBS (08/2017–04/2019, London):**
  - Designed real-time pipelines using **graph analytics** and **process mining** to surface organisational insights and support decision automation.
  - Developed observability tooling with **Kafka**, **Elasticsearch**, and Python, connecting data infrastructure to downstream analytics and business-facing use cases.

**Roles in UK Institutions** 2010–2016

**Assoc. Lect. @GSM | Coding Teacher @DGC | TA @KCL|UCL**

Taught a broad range of computing subjects during doctoral studies, covering both undergraduate and pre-university curricula. The through-line from this period is deep technical communication, strong fundamentals, and the ability to make complex ideas operational for others.

- **2012–2016:** Lectured in Computing, MATLAB, Java, Python, HTML, CSS at GSM and David Game College
- **2010–2014:** TA at King's College London (KCL) and UCL, covering Java, Data Structures, Systems Architecture, Semantic Web, SQL, and AI.

## education

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2010–2014 | <b>Ph.D. in Computer Science</b><br><i>Thesis: Persuasion Dialogues &amp; Opponent Modelling.</i><br>Developed logical inference methods for large <b>Knowledge Graphs</b> , combining symbolic reasoning, <b>graph analytics</b> , and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received <b>Best Poster Award</b> at IJCAI 2013 (413 submissions). Awarded <b>EPSRC PhD Scholarship</b> , <b>Outstanding TA Award</b> (student-nominated), and earned the <b>Graduate Certificate in Academic Practice</b> .<br><a href="#">Access Publications on Google Scholar</a> | King's College London, UK  |
| 2004–2010 | <b>Diploma (BEng) in Computer Engineering</b><br>Five-year polytechnic degree in Computer Science and Engineering with a <b>strong focus on mathematics</b> (10 courses), covering theoretical aspects, networking, and telecommunications. <b>Majored in Artificial Intelligence</b> .                                                                                                                                                                                                                                                                                                                                                   | University of Thessaly, GR |
| 2002–2004 | <b>Military Officer: Second Lieutenant</b><br>Obligatory army service; position acquired through <b>competitive IQ</b> and athletic exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | National Guard, CY         |

## committees/affiliations

|               |                                                                                                                                                                            |                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| since 10/2024 | <b>University College London, Dept. of Information Studies</b><br>Support AI application, standardisation, and ethics initiatives in industry and academia.                | Associate Researcher               |
| since 01/2021 | <b>ISO - International Organisation for Standardisation</b><br>Member of JTC'21 Working Group 3, developing AI standards aligned with EU policies and international norms. | Committee Expert Member for JTC'21 |

## talks/interviews

|      |                                                                                                                                                                   |                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 2023 | <b>Bill Raymonds, Agile in Action</b><br><b>Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa</b> , <a href="#">Details: YouTube</a> | Podcast Interview              |
| 2022 | <b>Open Data Science Conference</b><br><b>Dynamicio</b> , a python library for abstracting your I/O, <a href="#">Details: here</a>                                | Industry Conference Talk       |
| 2020 | <b>iunera: Big Data Innovators</b><br>The Agile approach in data science explained by an ML expert, <a href="#">Details: here</a>                                 | Interview Blog Post            |
| 2020 | <b>Big Data Warsaw</b><br>Monitoring and Analysing Communication and Trade Events as Graphs, <a href="#">Details: here</a>                                        | Industry Conference Talk       |
| 2018 | <b>Connected Data London</b><br>AI tribes and interconnections: what's graph got to do with it?, <a href="#">Details: here</a>                                    | Panelist @ Industry Conference |
| 2018 | <b>Minds Mastering Machines</b><br>A talk on doing Data Science the Agile Way, <a href="#">Details: here</a>                                                      | Industry Conference Talk       |

## certificates & awards

|         |                                                                                          |                                   |
|---------|------------------------------------------------------------------------------------------|-----------------------------------|
| 07/2020 | <b>AWS Certified Machine Learning - Specialty</b><br>Validation Number: D0R0WC7CN1VQ13SV | AWS                               |
| 01/2019 | <b>Process Mining</b><br>License number: AFU2VTP6273V                                    | Coursera Course Certificates      |
| 10/2017 | <b>Google Certified Professional - Data Engineer</b><br>Credential ID: 000000400         | Google Cloud Certified            |
| 07/2017 | <b>Advanced Elasticsearch: Data modelling</b><br>Enrolment Id: 33442                     | Elastic                           |
| 06/2016 | <b>Graph Analytics for Big Data</b><br>License number: NZADYAQ7N8FW                      | Coursera Course Certificates      |
| 05/2016 | <b>Apache Spark Developer Certification</b><br>License number: 1.x-0548                  | O'Reilly Media                    |
| 04/2016 | <b>Neo4j Certified Professional</b><br>License number: here                              | Neo Technology, Creators of Neo4j |